

DMPFC Prioritizes Social Consolidation at Rest

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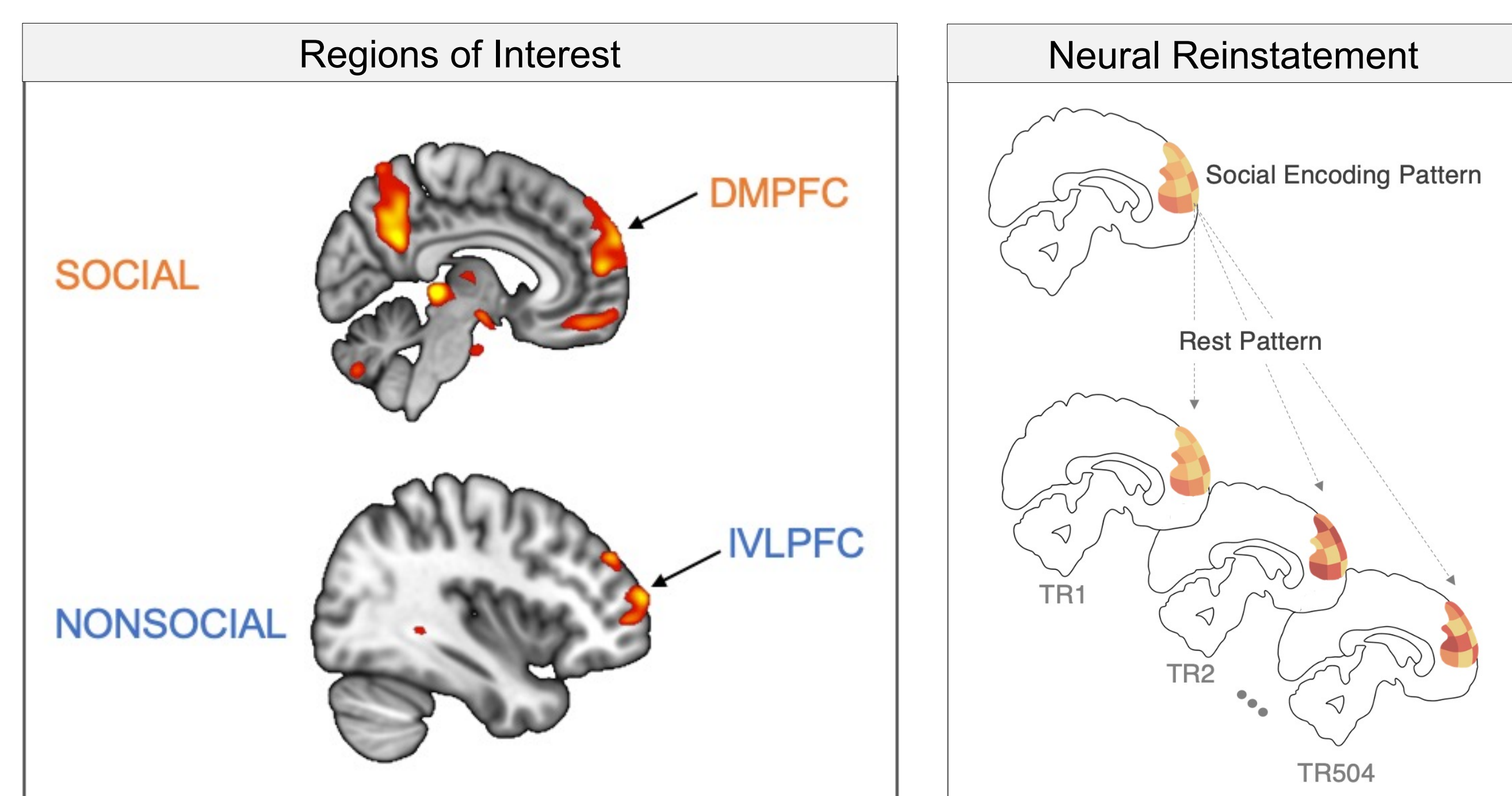
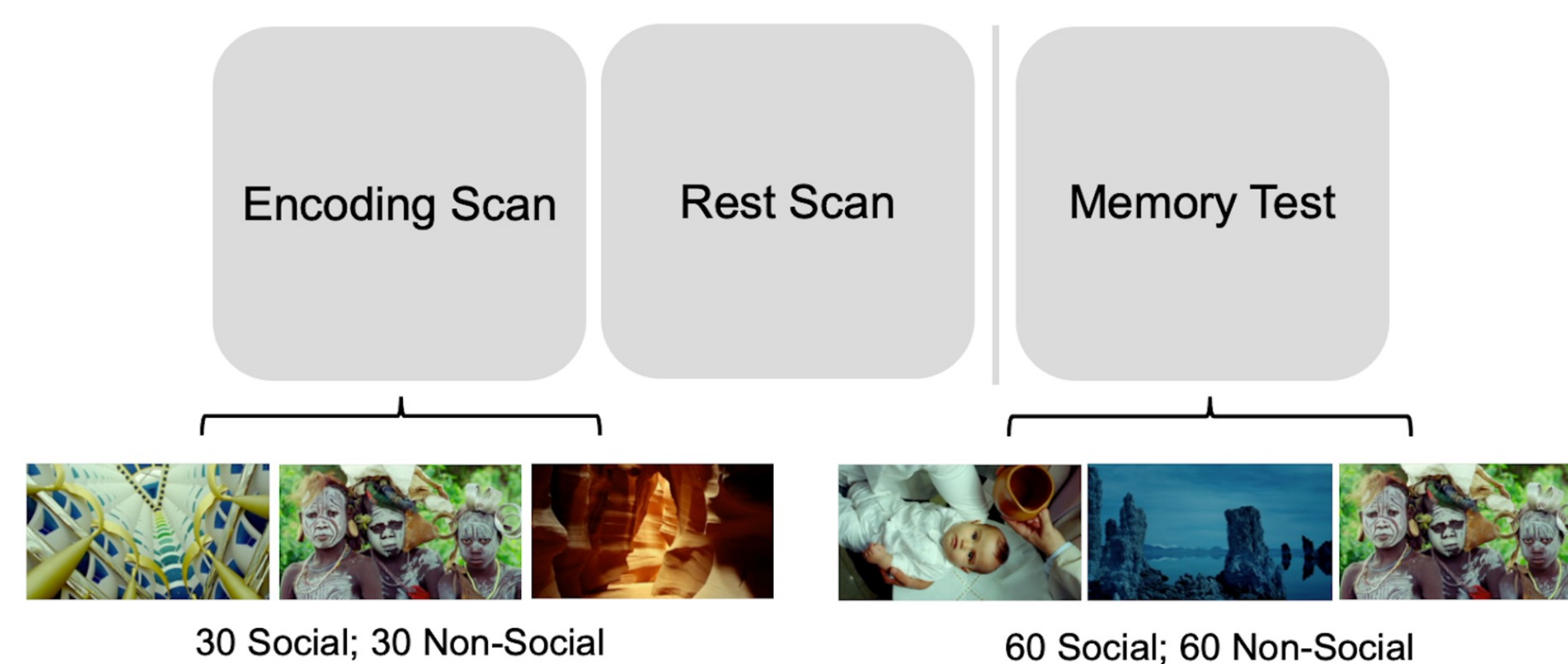
Introduction

In everyday life we encounter social and non-social information simultaneously. Yet, there is evidence that memory performance for social information is consistently better than memory performance for non-social information.

We hypothesize that:

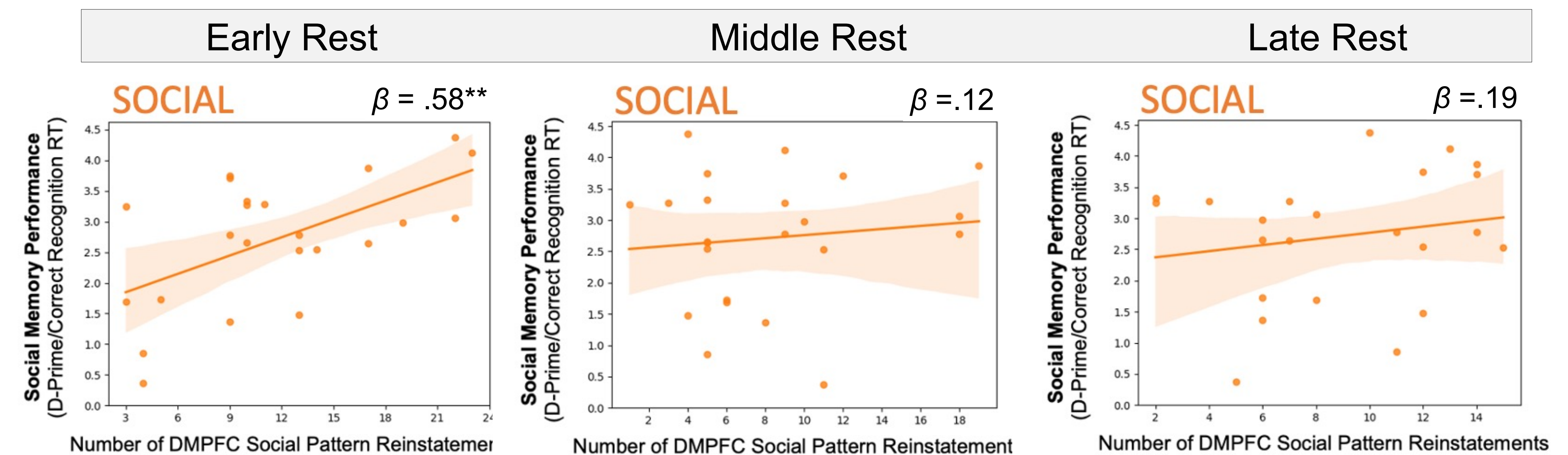
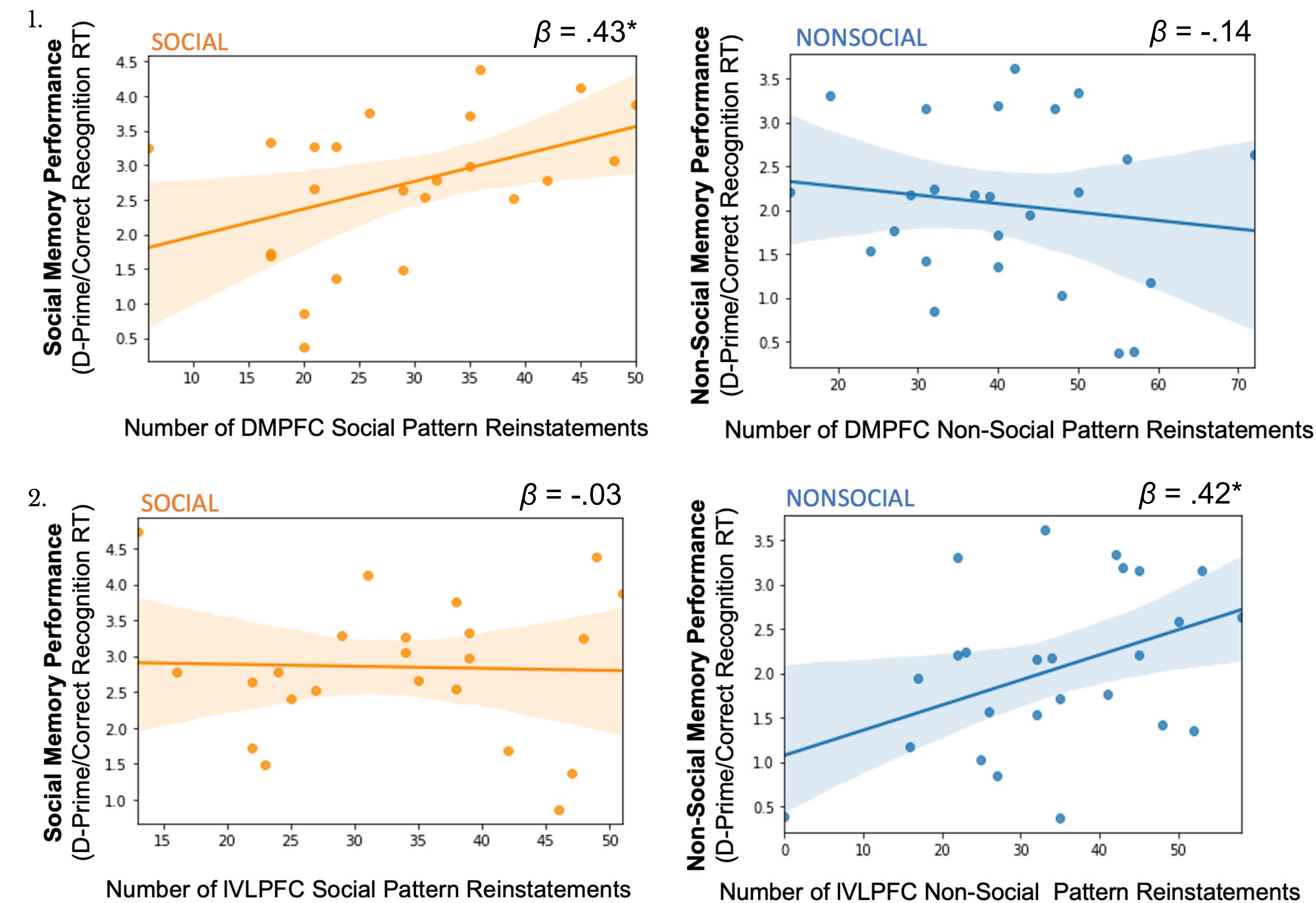
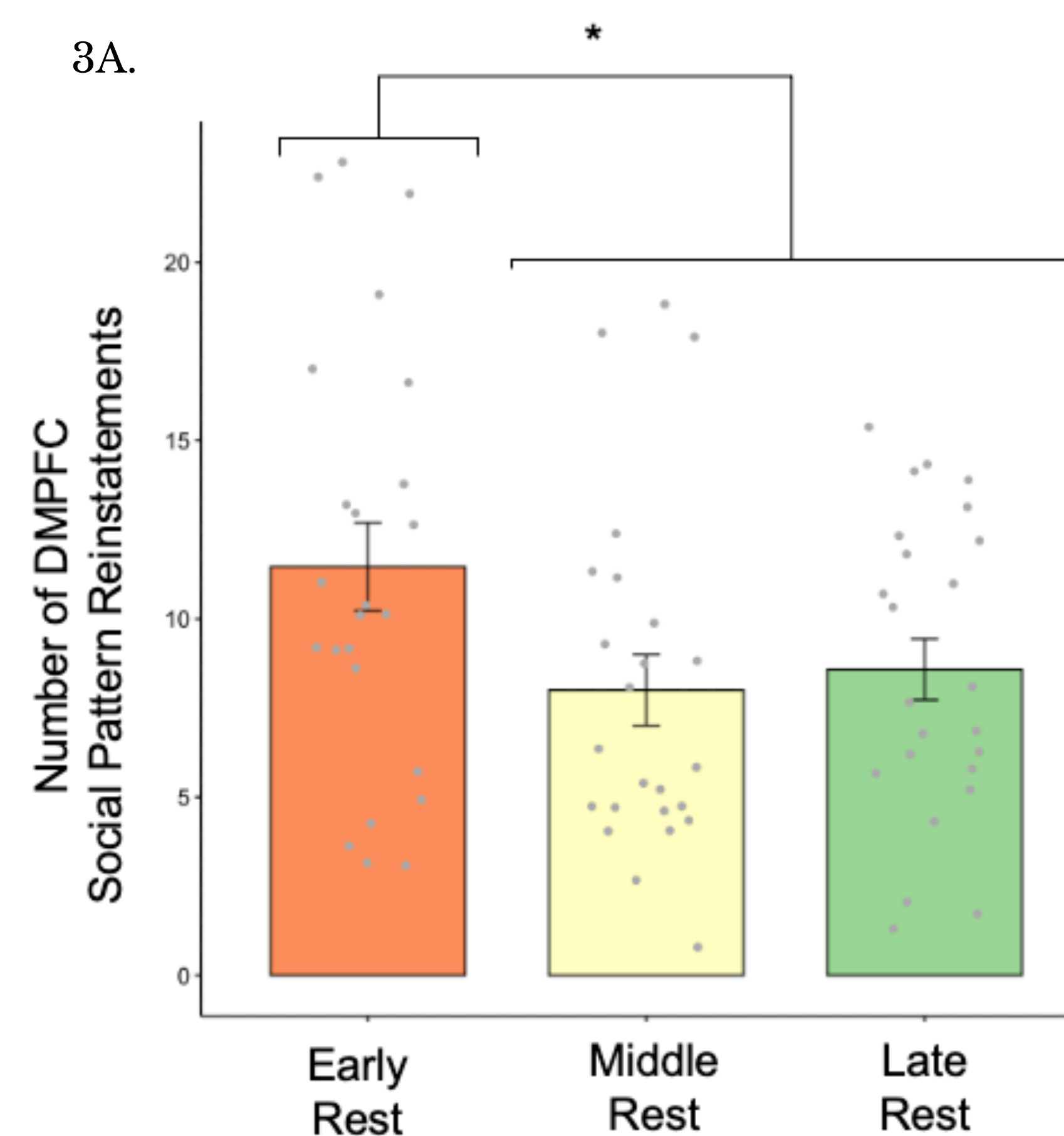
- The DMPFC, given its role in social cognition, may preferentially consolidate social information at rest¹.
- The VLPFC, given its role in nonsocial memory, may preferentially consolidate nonsocial information at rest².
- The brain may prioritize the consolidation of social information via early temporal dynamics at rest³.

Methods



Results

1. Social memory performance is positively correlated with reinstatement of social items in the DMPFC.
 2. Non-social memory performance is positively correlated with reinstatement of non-social items in the VLPFC.
- 3A. Social consolidation in the DMPFC is specific to early rest compared to middle and late rest.
- 3B. Nonsocial consolidation in the VLPFC is not specific to early rest when compared with middle and late rest.

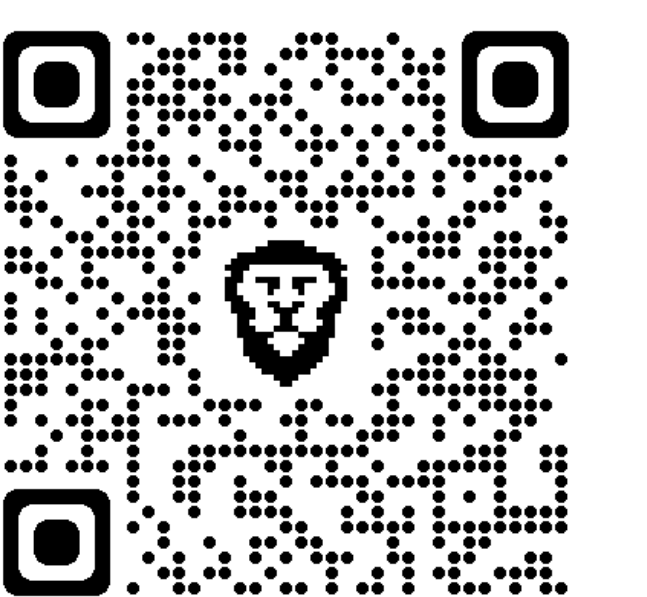


Conclusion

- The DMPFC - a key node of the default network - prioritizes the consolidation of social information during rest.
- The findings provide key support for the suggestion that the human brain may be “social by default.”

References

1. Meyer, M. L., et al. (2019). Evidence That Default Network Connectivity During Rest Consolidates Social Information. *Cerebral cortex (New York, N.Y. : 1991)*, 29(5), 1910–1920.
2. Badre, D. & Wagner, A. D. (2007). Left ventrolateral prefrontal cortex and the cognitive control of memory. *Neuropsychologia*, 45(13), 2883–2901.
3. Schapiro, A. C., et al. (2018). Human hippocampal replay during rest prioritizes weakly learned information and predicts memory performance. *Nature Communications*, 9(1), 3920.



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